Applied Immunology

Science

May, 2026

setu.ie

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Applied Immunology (A14139)

Short Title:	Applied Immunology		
Faculty	Science and Computing		
Department	Science		
Cluster	Biology		
Author	EMCNEELA		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

The aims of this module are to provide students with detailed information on the biology of the immune system in health and disease, and to provide insight into the applications of immunoanalytical techniques in industrial, clinical and research environments. The module aims to equip students with the skills necessary to understand the importance of immunology, underpinning disease prevention, diagnostics and therapeutics. From a practical perspective, this module will familiarise students with several different immunoanalytical techniques used routinely to measure and characterise immune responses.

Programmes

`BIOL-0009`	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`)	3 / 5 / M
`BIOL-0009`	BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`)	3 / 5 / M

Indicative Content

- Overview of the immune system; cells and organs of the immune system; innate and adaptive immunity
- Innate immunity: receptors and activation; complement; inflammation
- Adaptive immunity: T cells; B cells; MHC and antigen presentation; immunological memory; tolerance; interactions between innate and adaptive immunity
- Antibodies: structure; subclasses; specificity and diversity; effector functions
- Antigens: hapten; immunogenicity and epitopes
- Therapeutic antibody engineering and manufacture: polyclonal; monoclonal; antibody fragments; chimeric; humanised and fully human antibodies
- Immunoanalysis: agglutination; precipitation; enzyme immunoassays; flow cytometry; new developments; research and clinical applications
- Tumour immunology: tumours and metastasis; tumour antigens; immune responses and immune evasion; immunotherapies
- Vaccine technology: traditional and new vaccine technologies

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Communicate the fundamental mechanisms underlying infection, immunity and inflammation.
2. Compare and contrast the cells and protective mechanisms of the innate and adaptive immune systems.
3. Illustrate the general structural features of antibodies and relate these to their functions in adaptive immunity and in immunotherapies.
4. Explore a range of immunological techniques and interpret the data obtained.
5. Explore a range of traditional and newer generation immunoassays for detection of cellular and humoral immune responses.
6. Appraise new and emerging cancer immunotherapies in relation to their enhancing effects on the immune system and their interactions with the tumour microenvironment.

Learning and Teaching Methods

- Lectures.
- Laboratory-based practicals.
- Self-directed study.
- This module will use a blended teaching approach combining face-to-face lectures with in-class discussions and e-activities and videos delivered through the Moodle virtual learning environment.
- Formative feedback will be provided throughout to develop critical, independent learning.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	60%		1,2,3,5,6
Continuous Assessment	40%		
Practical		35%	4
Assignment		5%	1

Assessment Criteria

<40%: No understanding of applied immunology. Unable to carry out critical thinking or practical skills.

40%-49%: Demonstrate a limited knowledge of applied immunology. May have some difficulty with critical thinking and problem solving. Adequate practical skills.

50%-59%: A good understanding of applied immunology with demonstration of some critical thinking and problem solving skills. Good practical skills.

60%-69%: In addition to the above, good critical thinking skills and showing a high level of competence and efficiency in theoretical and practical components.

70%-100%: A highly developed and extensive understanding of applied immunology. Excellent technical and data analysis ability in carrying out and reporting on practical tasks.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	30	
Independent Learning	69	

Essential Material

- Delves, P.J., S.J. Martin, D.R. Burton and I.M. Roitt. _Roitt's Essential Immunology_. Chichester, West Sussex; Hoboken, N.J.: Wiley-Blackwell, 2011.

Supplementary Material

- "British Society for Immunology." https://www.immunology.org/
- "Immunobiology NCBI." http://www.ncbi.nlm.nih.gov/books/NBK10757/
- "Roitt's Essential Immunology Resources." http://www.roitt.com/
- "Scientific Article Search Engine." http://www.ncbi.nlm.nih.gov/pubmed/
- Coico, R., G. Sunshine and E. Benjamini. _Immunology: A Short Course_. 5th ed. Hoboken, N.J.: Wiley-Liss, 2003.
- Murphy, K. _Janeway's Immunobiology_. 8th ed. New York: Garland Science - Taylor and Francis Group, 2014.
- Owen, J., J. Punt and S. Stranford. _Kuby Immunology_. 7th ed. New York: W H Freeman and Co., 2013.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab